



MODULE 5

Planning & Multi-Agent Systems

From Single Agents to Collaborative Teams — Planning, ReAct, Orchestration

What are Multi-Agent Systems?



Multi-agent systems use multiple specialized AI agents, each with distinct roles, tools, and expertise, working together to accomplish tasks too complex for any single agent. Like a team of specialists instead of one generalist.



Specialization

Each agent masters one domain — deeper expertise



Parallelism

Independent agents work simultaneously — faster



Modularity

Add/remove agents without rewriting the whole system



Reliability

If one agent fails, others continue — graceful degradation

Role-Based Agent Design

Each agent has a defined role, goal, backstory, and tool access — shaping its behavior and decision-making.

Planner Agent

Decompose user goal into sub-tasks, create execution plan, assign tasks to other agents

None (reasoning only)

Researcher Agent

Gather information from web search, databases, and documents. Produce fact-based summaries

web_search, web_fetch, read_file

Analyst Agent

Analyze data, calculate metrics, identify patterns and insights from gathered information

calculator, code_exec, query_db

Writer Agent

Synthesize research and analysis into clear, well-structured reports and documents

create_document, format_text

Reviewer Agent

Quality-check outputs for accuracy, completeness, and adherence to requirements

None (reasoning + critique)

Agent Collaboration Models

Sequential Chain:



Hierarchical (Supervisor):



Debate / Discussion:



<https://arxiv.org/pdf/2512.04668>

Agent Communication Patterns

Message Passing

Agents send structured messages to each other. Each message includes sender, content, and metadata. Most common.

Shared State

All agents read/write to a shared state object. Good for coordination but can cause conflicts.

Event-Driven

Agents publish events to a bus. Other agents subscribe and react. Loose coupling, scalable.

Blackboard

Central knowledge store. Agents read problems, write solutions. Others build on partial solutions.

Direct Handoff

Agent A calls Agent B directly, waits for result, continues. Simple but tightly coupled.

SECTION 5

Orchestration Patterns

How to coordinate agent teams for complex workflows

Pattern 1: Sequential Workflow

Agents execute in a fixed order — each agent's output feeds the next. Like an assembly line.

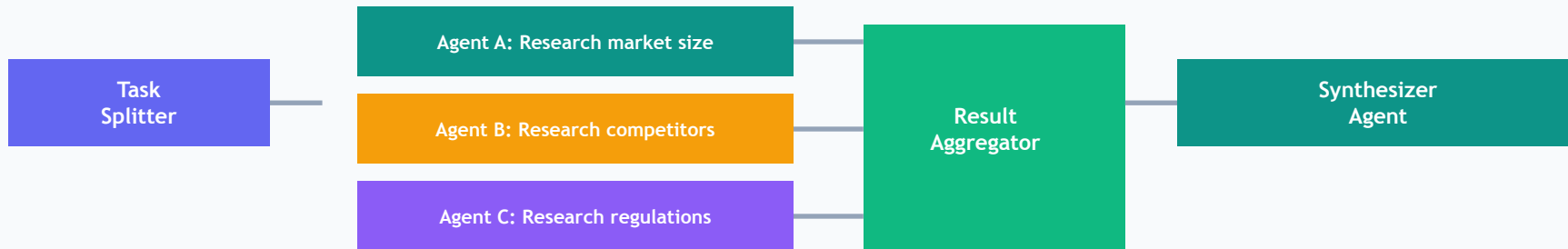


Example: Content Creation Pipeline

1. Researcher: Search web for topic → collects 10 sources, extracts key points
2. Analyst: Identify trends, compare perspectives, find unique angles
3. Writer: Create 2000-word article with citations, structure, and narrative
4. Reviewer: Check factual accuracy, grammar, completeness → approve or revise
5. If revision needed → Writer gets feedback → rewrites → Reviewer re-checks

Pattern 2: Parallel Agents

Independent agents run simultaneously, then results are aggregated. Faster for independent sub-tasks.



When to Use Parallel:

Independent research: each agent searches a different topic/region/source

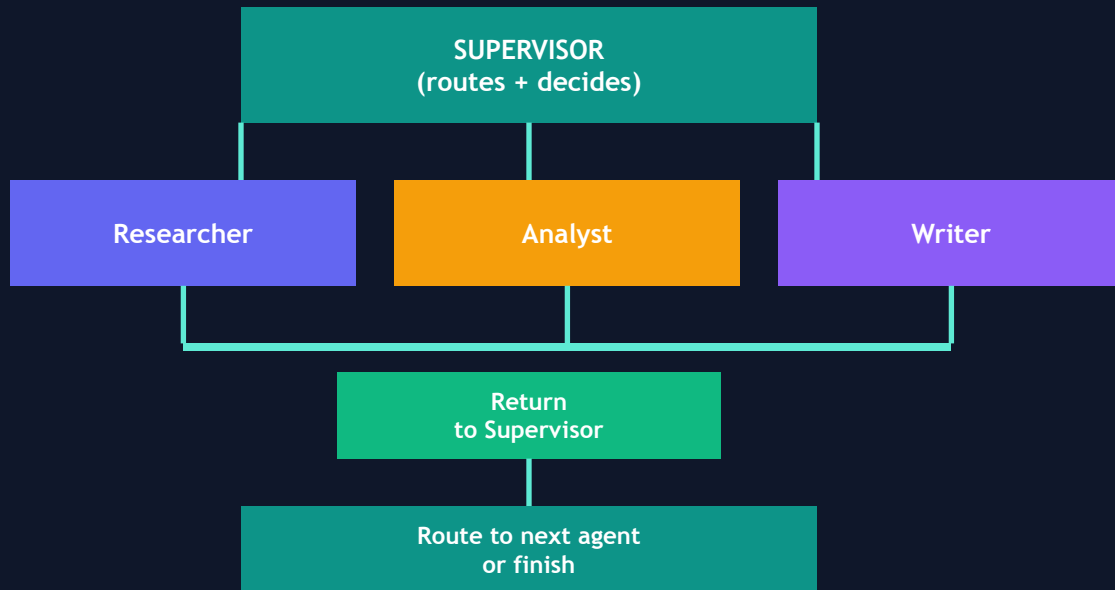
Multi-perspective analysis: agents take different viewpoints, then a synthesizer combines

Batch processing: process 10 documents simultaneously, one agent per document

Speedup: 3 agents in parallel = ~3x faster than sequential (minus coordination overhead)

Pattern 3: Supervisor / Manager

A supervisor agent delegates tasks, monitors progress, and makes routing decisions dynamically.



- **Dynamic Routing:** Supervisor reads results, decides which agent goes next — not a fixed sequence
- **Quality Gates:** Supervisor checks each output before routing — rejects poor work back to the agent
- **Human-in-Loop:** Supervisor can escalate to human for approval at critical decision points

Orchestration Patterns: Comparison

Pattern	Coordination	Flexibility	Best For
Sequential	Fixed order	Low	Pipelines, content creation
Parallel	Independent	Medium	Independent research
Supervisor	Dynamic routing	High	Complex, conditional tasks
Debate	Multi-perspective	High	Analysis, decision-making
Swarm	Agent handoffs	Very high	Customer support, routing
Hierarchical	Tree delegation	High	Enterprise workflows

When to Use Multi-Agent

Simple Q&A, lookups?

Single agent — multi-agent is overkill for simple tasks

Multi-step with dependencies?

Single agent with ReAct — one agent can handle 5-10 step tasks

Requires different expertise?

Multi-agent — research needs search skills, analysis needs math skills

Independent parallel work?

Multi-agent parallel — 3 agents researching simultaneously

Dynamic routing needed?

Supervisor pattern — manager decides who handles each part

Customer support flows?

Swarm pattern — triage → specialist → resolution

Human-in-the-Loop Patterns



Plan Approval

Agent creates a plan → human reviews and approves before execution begins. Best for high-stakes tasks.

Checkpoint Reviews

Agent pauses at defined checkpoints → human inspects intermediate output → continues or revises.

Escalation Handling

Agent detects low confidence or ambiguity → escalates to human with context and options.

Quality Gates

Final output reviewed by human before delivery. Agent handles 90% autonomously, human checks 10%.

Feedback Loop

Human rates agent outputs. Ratings feed back to improve prompts and agent behavior over time.



Thank You!!!

